Dr. Eng. Paulina Stempin

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RESEARCH INTERESTS

- Nonlocal mechanics; fractional continuum mechanics; scale effect; multiscale mechanics
- Computational Mechanics, finite element methods
- Mathematical modelling

POSITIONS

2024/11/01 – present	**Assistant Professor** Institute of Structural Analysis Poznan University of Technology, Poland
2021/11/01 – 2024/10/31	**Research Assistant** Institute of Structural Analysis Poznan University of Technology, Poland

EDUCATION

2024/03/26	**Dr. Eng.** in Civil Engineering, Geodesy and Transport Poznan University of Technology, Poland *Structural models in the framework of space-Fractional Continuum Mechanics* Supervisor: Prof. Wojciech Sumelka
2019/07/04	**M.Sc. Eng.** in Civil Engineering: Structural Engineering Poznan University of Technology, Poland *Numerical analysis of the acoustic screen* (in Polish) Supervisor: Prof. Wojciech Sumelka
2018/02/07	**B.Sc. Eng.** in Civil Engineering Poznan University of Technology, Poland *Structural design of arena* (in Polish) Supervisor: Dr. Eng. Janusz Debinski

RESEARCH GRANTS

03/04/2023 – 02/04/2027 **Principal Investigator in project 2022/45/N/ST8/02421**
Mechanics of bar-and-plate structures with strong scale effects – mathematical modelling and experimental analysis
Funder: National Science Center, Poland

01/03/2024 – 28/02/2026 **Contractor in project PM-II/SP/0077/2024/02**
Development of a Reference Setup for Testing the Dynamic Mechanical Properties of Materials
Funder: MNiSW - Polska Metrologia II, Poland
Leader: Prof. Wojciech Sumelka

01/12/2019 – 24/11/2021 **Scholarship in project 2017/27/B/ST8/00351**
Development of fractional continuum mechanical models
Funder: National Science Center, Poland
Leader: Prof. Wojciech Sumelka

PUBLICATIONS

Research metrics (Scopus, Aug 25 2026): H-index: 6; citations: 151 (119 without self-citations); summed IF = 60.968.
* Corresponding author

13. **Paulina Stempin***, Wojciech Sumelka. (2026). *B-spline-enhanced space-fractional finite element method with application to limit state analysis of 3D scale-sensitive truss structures*. Computational Mechanics, vol. ., p. .–. *IF = 4.500*

12. **Paulina Stempin***, Wojciech Sumelka. (2026). *B-spline-based space-fractional finite element formulation for planar scale-sensitive Timoshenko frames*. Computers & Structures, vol. 330, p. 108342-1-108342-16 *IF = 5.600*

11. Kevin Moj, Grzegorz Robak, Joanna Malecka, Jolanta Królczyk, Marcin Nowak, **Paulina Stempin**, Kamil Cybul, Michał Wieczorowski, Wojciech Sumelka, Tomasz Jankowiak. (2026). *Statistical representativeness of the microstructure and surface topography in steel reference samples in a dynamic compression test*. Journal of Materials Research and Technology, vol. 42, p. 9270-9286 *IF = 7.300*

10. Krzysztof Szajek, **Paulina Stempin**, Wojciech Sumelka. (2025). *Comprehensive analysis of complexity of damage identification in micro-beam like structures based on silver nanowires experimental data*. Engineering with Computers, vol. 41, p. 3719–3736 *IF = 4.100*

9. **Paulina Stempin***, Wojciech Sumelka. (2025). *Space-Fractional Finite Element Method for Scale Sensitive Truss Structures - Theory and Implementation*. Computational Mechanics, vol. 76, p. 905–922 *IF = 4.500*

8. **Paulina Stempin**, Wojciech Sumelka. (2025). *Approximation of Fractional Caputo Derivative of Variable Order and Variable Terminals with Application to Initial/Boundary Value Problems.*. Fractal and Fractional, vol. 9, iss. 5, p. 269-1-269-17 *IF = 3.500*

7. Krzysztof Szajek, **Paulina Stempin**, Wojciech Sumelka. (2024). *Damage identification in scale sensitive beam-like nano-joints in the framework of space-fractional Euler–Bernoulli theory*. Engineering Failure Analysis, vol. 164, p. 108606-1-108606-20 *IF = 5.700*

6. **Paulina Stempin**, Tomasz P. Pawlak, Wojciech Sumelka. (2023). *Formulation of non-local space-fractional plate model and validation for composite micro-plates*. International Journal of Engineering Science, vol. 192, p. 103932-1-103932-17 *IF = 5.700*

5. **Paulina Stempin**, Wojciech Sumelka. (2022). *Space-fractional small-strain plasticity model for microbeams including grain size effect*. International Journal of Engineering Science, vol. 175, p. 103672-1-103672-12 — *IF = 6.600*

4. **Paulina Stempin**, Wojciech Sumelka. (2021). *Dynamics of Space-Fractional Euler-Bernoulli and Timoshenko Beams*. Materials, vol. 14, no. 8, p. 1817-1-1817-19 — *IF = 3.748*

3. **Paulina Stempin**, Wojciech Sumelka. (2021). *Formulation and experimental validation of space-fractional Timoshenko beam model with functionally graded materials effects*. Computational Mechanics, vol. 68, iss. 3, p. 697-708 — *IF = 4.391*

2. **Paulina Stempin**, Wojciech Sumelka. (2020). *Space-fractional Euler-Bernoulli beam model – Theory and identification for silver nanobeam bending*. International Journal of Mechanical Sciences, vol. 186, p. 105902-1-105902-7 — *IF = 5.329*

1. **Paulina Stempin**, Wojciech Sumelka. (2020). *Numerical Analysis of Road Acoustic Screen*. Archives of Civil Engineering, vol. 66, no. 2, p. 191-210

CONFERENCES - SELECTED PRESENTATIONS

9. **(Keynote Lecture): Paulina Stempin**, Wojciech Sumelka. *Space-Fractional Finite Element Formulation for 3D Nonlocal Frames.* 17th World Congress on Computational Mechanics & 10th European Congress on Computational Methods in Applied Sciences and Engineering (**WCCM-ECCOMAS 2026**), July 19-24, 2026, Munich, Germany

8. **Paulina Stempin**, Wojciech Sumelka. *Finite Element Method for Space-Fractional Truss and Frame.* 8th edition of the ECCOMAS Young Investigators Conference (**YIC2025**), September 17-19, 2025, Pescara, Italy

7. **Paulina Stempin**, Wojciech Sumelka. *Space-fractional finite element approach for size-dependent frame structures.* 26th International Conference on Computer Methods in Mechanics (**CMM2025**), July 8-11, 2025, Łódź, Poland

6. **Paulina Stempin**, Wojciech Sumelka. *Development of space-fractional finite element for scale-sensitive truss structures.* 95th Annual Meeting of the International Association of Applied Mathematics and Mechanics (**GAMM2025**), April 7-11, 2025, Poznań, Poland

5. **Paulina Stempin**, Wojciech Sumelka. *Modelling of Nonlocal Truss Structures in the Framework of Space-Fractional Continuum Mechanics.* 9th European Congress on Computational Methods in Applied Sciences and Engineering (**ECCOMAS CONGRESS 2024**), June 2-8, 2024, Lisbon, Portugal

4. **Paulina Stempin**, Krzysztof Szajek, Wojciech Sumelka. *Space-fractional continuum mechanics as a method of capturing the scale effect.* 19th European Mechanics of Materials Conferences (**EMMC 2024**), May 29-31, 2024, Madrid, Spain

3. **Paulina Stempin**, Wojciech Sumelka. *Modelling of nonlocal thick plates in the framework of space-fractional continuum mechanics.* 5th Polish Congress of Mechanics & 25th International Conference on Computer Methods in Mechanics (**PCM-CMM 2023**), September 4-7, 2023, Gliwice, Poland

2. **Paulina Stempin**, Wojciech Sumelka. *Plastic Hinge Formation in the Framework of the Space-Fractional Beam Theory.* 15th World Congress on Computation Mechanics & 8th Asian Pacific Congress on Computation Mechanics (**WCCM-APCOM 2022**), July 31 – August 5, 2022, Yokohama, Japan (Virtual Congress)

1. **Paulina Stempin**, Wojciech Sumelka. *Selected beam theories in the framework of space-fractional mechanics.* 14th World Congress in Computational Mechanics and ECCOMAS Congress (**WCCM-ECCOMAS Virtual Congress**), January 11-15, 2021, Paris, France (Virtual Congress)

AWARDS

11/07/2025	**Third place in Jan Szmelter Award** for the best paper presented by the young researcher 26th International Conference on Computer Methods in Mechanics, July 8-11, 2025, Łódź, Poland
10/11/2023	**Distinction for the preparation and presentation of a scientific paper by young scientists (pre-doctoral)** XIII Seminarium Naukowe ZINTEGROWANE STUDIA PODSTAW DEFORMACJI PLASTYCZNEJ METALI conference November 7-10, 2023, Łańcut, Poland
07/09/2023	**Third place in Jan Szmelter Award** 25th International Conference on Computer Methods in Mechanics & 5th Polish Congress of Mechanics, September 4-7, 2023, Gliwice, Poland
2015 to 2019	**Scholarship** of the Rector of Poznan University of Technology for outstanding study results (the top 10% of students in the academic year)

COURSE COMPLETION

- *Training in operating the Inspect Micro L100 machine with LabMaster software*, training by ITA, October 16, 2025, Poznan, Poland
- 11th CISM–AIMETA Advanced Course on "*Machine Learning for Solid Mechanics*", course by CISM (International Centre for Mechanical Sciences), September 29 - October 03, 2025, Udine, Italy (Online Participation)
- *Short Course on Implementation of the DPG Method in a FE Code Supporting H1, H(curl), H(div), and L2 - Conforming Finite Elements*, course by Prof. Leszek Demkowicz and Dr. Stefan Henneking, June 14-15, 2022, Poznań, Poland

MEMBERSHIP

2023/10 to present	ECCOMAS Young Investigators Committee (EYIC)
2022/11/21 to present	Polish Association for Computational Mechanics
2026/07/19-24	Member of Next Generation Committee - WCCM-ECCOMAS 2026 (17th World Congress on Computational Mechanics & 10th European Congress on Computational Methods in Applied Sciences and Engineering), July 19-24, 2026, Munich, Germany

TEACHING EXPERIENCE

- Courses taught (2020-2026, Poznan University of Technology): Computational Mechanics; Computer aided design; Theory of structures; Numerical methods; Engineering graphics and CAD.
- Student Supervision: BSc theses – 3 completed; MSc theses – 1 ongoing; PhD theses (auxiliary supervisor) – 1 ongoing.

TECHNICAL STRENGTHS

Language	Polish (native), English (B2)
Programming language	Julia, Python, Scilab
Engineering software	Abaqus, Autodesk Robot Structural Analysis, AutoCAD